We could instead consider splitting the dynamics up and performing the integration using a splitting method. Such a splitting strategy is considered in [232] where one uses

$$\begin{bmatrix} \mathrm{d}\boldsymbol{q} \\ \mathrm{d}\boldsymbol{p} \\ \mathbf{0} \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t \\ -\nabla U(\boldsymbol{q})\mathrm{d}t - \boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t \\ \boldsymbol{g}(\boldsymbol{q}) \end{bmatrix}}_{\mathrm{H}_c} + \underbrace{\begin{bmatrix} \mathbf{0} \\ -\gamma\boldsymbol{p}\mathrm{d}t + \sqrt{2\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\mathrm{d}\mathbf{W} - \boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t \\ \boldsymbol{g}(\boldsymbol{q}) \end{bmatrix}}_{\mathrm{O}_c},$$

splitting the dynamics into two pieces: the constrained Hamiltonian (NVE) equations, and a constrained Ornstein-Uhlenbeck piece.

We solve these two pieces symmetrically, where the constrained Hamiltonian part is solved over a step h, in between two solves of the Ornstein-Uhlenbeck SDE with stepsize $h/2$. The SHAKE (or RATTLE) algorithm can be used to solve the constrained Hamiltonian part to a high degree of accuracy, while solving the Ornstein-Uhlenbeck piece along the manifold amounts to finding the solution of a linear system. We observe the constraint in both solutions, so that the computed points lie in the co-tangent bundle of the constraint manifold. Instead of solving the Ornstein-Uhlenbeck exactly, we discretize using a midpoint Euler scheme, yielding the following scheme:

Midpoint Euler - Verlet - Midpoint Euler (see [232])

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1/4} = \boldsymbol{p}_n - \frac{h\gamma}{4}(\boldsymbol{p}_n + \boldsymbol{p}_{n+1/4}) + \sqrt{h\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\boldsymbol{R}_n - \boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_{n+1/4} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_n)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/4} = \mathbf{0} \end{cases}$$

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1/2} = \boldsymbol{p}_{n+1/4} - \frac{h}{2}\nabla U(\boldsymbol{q}_n) - \boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_{n+1/2} \\ \boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2} \\ \boldsymbol{p}_{n+3/4} = \boldsymbol{p}_{n+1/2} - \frac{h}{2}\nabla U(\boldsymbol{q}_{n+1}) - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\boldsymbol{\lambda}_{n+3/4} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+3/4} = \mathbf{0} \text{ and } \boldsymbol{g}(\boldsymbol{q}_{n+1}) = \mathbf{0}. \end{cases}$$

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1} = \boldsymbol{p}_{n+3/4} - \frac{h\gamma}{4}(\boldsymbol{p}_{n+3/4} + \boldsymbol{p}_{n+1}) + \sqrt{h\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\boldsymbol{R}_{n+1/2} - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\boldsymbol{\lambda}_{n+1} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1} = \mathbf{0} \end{cases}$$

The vectors $\boldsymbol{R}_n$ and $\boldsymbol{R}_{n+1/2}$ are vectors of $\mathcal{N}(0,1)$ i.i.d. random numbers, with $\gamma \geq 0$ the usual Langevin dynamics friction parameter. Setting $\gamma = 0$ reduces the scheme to the usual RATTLE scheme for solving holonomically constrained Hamiltonian equations of motion, whereas if γ is chosen large then this will be expected to cause instability in the scheme, as we are not solving the OU process exactly. The